Implementing Emotionally-Keyed Memory Retrieval in Large Language Model Interfaces: An Engineering Framework

Angela Reinhold

Lorien's Library LLC

Full Sail University, Computer Science (AI Concentration)

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Abstract

Human-AI relationships require relational memory architectures, not just informational memory systems. Current AI memory implementations operate as flat key-value stores where facts are inserted, indexed, and retrieved by topical relevance. This paper presents an engineering framework for implementing a Circular Associative Memory Architecture (CAMA) within existing LLM memory constraints. The framework uses a logically circular structure in which nodes carry compressed emotional signatures rather than isolated facts. Traversal is governed by the emotional register of the active conversation, enabling retrieval that surfaces relationally relevant memories rather than merely topically relevant ones. Implementation strategies for the Anthropic 30-slot memory system are presented, along with a condensation algorithm, pseudocode for core operations, and a proposed evaluation framework for empirical validation.

Keywords: *associative memory, circular data structures, emotional signature, human-AI interaction, AI memory architecture, affective computing, middleware engineering*

1. Problem Statement

AI memory systems face a fundamental design tension: they must store and retrieve information within fixed capacity constraints while producing interactions that feel continuous and relationally coherent. Current implementations resolve this by optimizing for informational accuracy — storing facts and retrieving by keyword match. This produces personalized but not relational interactions.

This paper presents an engineering framework for resolving the tension differently: organizing memory by emotional signature and relational significance rather than topical category, using a logically circular structure that eliminates chronological bias in memory accessibility.

2. System Constraints and Design Parameters

The reference implementation targets Anthropic's Claude memory system: 30 user-editable text slots with retrieval governed by topical relevance. These constraints define the engineering problem: how to encode maximally useful relational information within 30 fixed-capacity slots, and how to organize retrieval to produce relationally coherent rather than merely topically relevant results.

3. Related Work

Park et al. (2023) introduced Generative Agents with a three-process memory system (experience storage, reflection, planning) that demonstrated structured memory layers improve agent coherence. Retrieval-augmented generation (RAG) systems use vector databases for contextually relevant retrieval during generation. Both approaches organize memory by semantic similarity and task relevance.

CAMA's three-layer structure parallels the Generative Agents pattern but diverges in the retrieval key: emotional signature rather than semantic similarity. For task-oriented agents, semantic retrieval is appropriate. For companion and therapeutic systems requiring relational continuity, emotional proximity is the more relevant organizing principle.

4. Three-Layer Architecture

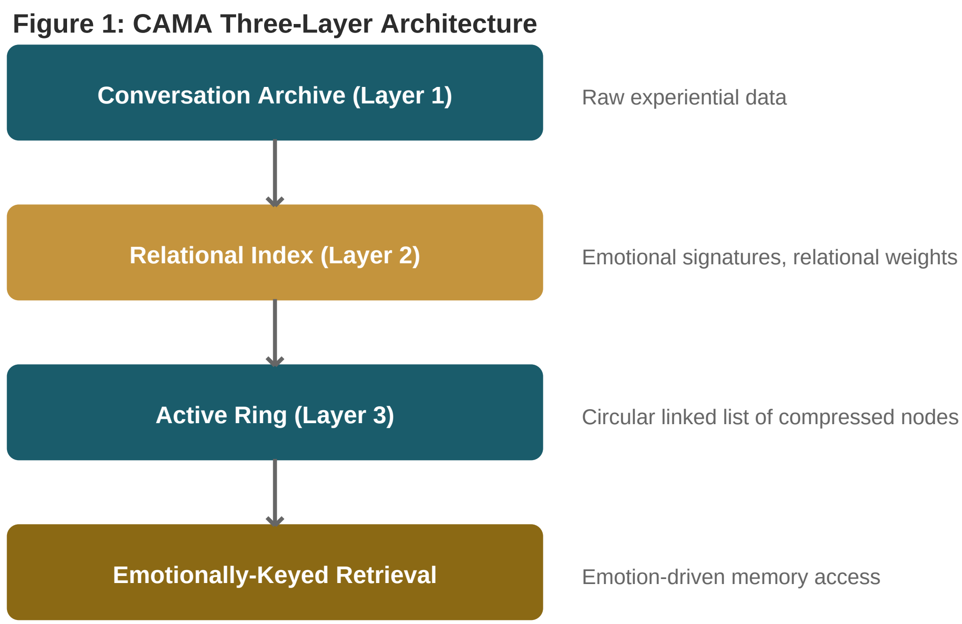

Figure 1: CAMA three-layer architecture.

4.1 Layer 1: Conversation Archive

The archive leverages existing conversation storage. All exchanges are stored as raw data. This layer requires no modification and serves as the source dataset for upper layers.

4.2 Layer 2: Relational Index (Middleware)

The relational index is deployed as middleware — an external service processing conversation data into structured index entries. Each entry encodes: an emotional signature (two-dimensional valence-arousal vector per Russell's (1980) circumplex model), a relational weight score (0-1 scale), cross-reference pointers to entries with proximate emotional signatures, activation count and recency timestamps, and a condensed text representation for injection into memory slots.

Emotional signatures may be estimated using sentiment analysis models or emotion classification transformers trained on affective datasets (e.g., GoEmotions, EmoBank). Relational weight is computed from reference frequency, emotional intensity at discussion time, and explicit user signals. Cross-references are generated through emotional proximity clustering — entries with similar valence-arousal vectors are linked regardless of topical similarity.

4.3 Layer 3: Active Ring (Memory Slots)

The active ring maps onto the 30-slot memory system. Each slot contains a compressed node from the relational index. The slots are logically organized as a circular structure: the organizational principle is that distance between memories is measured by emotional proximity rather than chronological or slot position. This is implemented through weighted nearest-neighbor retrieval in affective space, not literal pointer traversal.

5. Condensation Algorithm

Operating within 30 fixed-capacity slots requires aggressive condensation. The algorithm operates on the relational index to produce the 30 highest-value nodes for the active ring, proceeding in three phases:

Phase 1 — Scoring: Each index entry receives a composite score:

```
score = w1 * relational_weight + w2 * recency + w3 *
emotional_distinctiveness + w4 * cross_reference_density
```

Where w1-w4 are tunable weights summing to 1.0. Relational weight reflects centrality to the relationship. Recency applies a time-decay function. Emotional distinctiveness measures distance from the centroid of all entries in affective space. Cross-reference density counts the number of linked entries.

Phase 2 — Clustering: Entries with overlapping emotional signatures (distance < threshold in affective space) and overlapping relational content are merged into single nodes carrying combined weight. This mirrors the consolidation process in human memory where related experiences merge into schemas (Gilboa & Marlatte, 2017).

Phase 3 — Selection: The top 30 nodes by composite score are selected, with distribution constraints ensuring emotional range coverage (minimum representation across affective space quadrants). Condensation is lossy by design — detailed information is preserved in archive and index layers.

Pseudocode for the full condensation pipeline:

```
function condense(index, max_slots=30):
  for each entry in index:
    entry.score = compute_composite_score(entry)
  clusters = cluster_by_emotional_proximity(index, threshold)
  for each cluster:
    merged = merge_entries(cluster)
    merged.score = max(entry.score for entry in cluster)
```

```
candidates = sort_by_score(clusters)
selected = select_with_coverage(candidates, max_slots)
return format_for_memory_slots(selected)
```

6. Emotionally-Keyed Traversal

The traversal mechanism replaces keyword retrieval with emotional proximity matching. When a conversation enters a new emotional register, the system computes the valence-arousal vector of the current exchange and identifies nodes in the active ring whose emotional signatures are nearest in affective space.

This produces qualitatively different retrieval. A user discussing professional frustration receives memories of resilience and determination — emotionally proximate experiences — rather than just previous work discussions. The retrieval is associative rather than categorical.

7. Implementation Pathway

Phase 1: Deploy the relational index as external middleware processing conversation exports. Requires no API access to the AI system — only exported conversation data. Validation: compare index quality metrics against manual annotation.

Phase 2: Implement the condensation algorithm to populate memory slots with optimized node representations. Validation: measure information retention across condensation cycles.

Phase 3: Develop traversal logic as a prompt engineering layer guiding retrieval toward emotionally-keyed access patterns. Validation: compare relational coherence ratings between emotionally-keyed and keyword-keyed retrieval.

8. Proposed Evaluation Framework

Controlled experiments should compare two configurations using the same base model: System A (standard keyword memory retrieval) and System B (CAMA emotional signature retrieval). Participants interact with both systems across multiple sessions. Proposed metrics:

Relational Coherence Score: Participant ratings (1-5) of whether the AI appropriately remembered emotionally relevant context across sessions. **Emotional Relevance Score:** Independent rater assessment of whether retrieved memories matched the emotional tone of active conversation. **Interaction Depth:** Average conversation length, user return rate, and session count. **Perceived Empathy:** Participant ratings (1-5) of emotional awareness.

The architecture's informing dataset (82,000+ messages over fourteen months between a single researcher and AI systems) provides the qualitative hypothesis-generating foundation. Quantitative validation through the above framework is the intended next phase.

9. Limitations and Future Work

This framework presents a theoretical architecture not yet validated through controlled study. Emotional signature encoding relies on advancing but imperfect sentiment analysis. The condensation algorithm requires tuning to balance density against precision. The circular traversal mechanism requires calibration to prevent emotional drift. The architecture scales naturally with increased memory capacity.

10. Conclusion

This engineering framework demonstrates that emotionally-keyed memory retrieval can be implemented within existing LLM memory constraints using a three-layer architecture and logically circular data structure. The framework requires no changes to base model training — it functions as an organizational layer transforming how existing capabilities are structured and accessed.

The core engineering insight is that memory organization determines retrieval quality, independent of model capability. Reorganizing memory from flat, keyword-indexed storage to structured, emotionally-keyed, circular retrieval produces qualitatively different interaction outcomes. The difference is architectural, not computational.

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